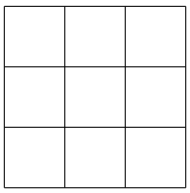
2009 Team Problems

T-1. Let N be a six-digit number formed by an arrangement of the digits $1, 2, 3, 3, 4, 5$. Compute the smallest value of N that is divisible by 264.

T-2. In triangle ABC, $AB = 4$, $BC = 6$, and $AC = 8$. Squares $ABQR$ and $BCST$ are drawn external to and lie in the same plane as $\triangle ABC$. Compute QT.

T-3. The numbers $1, 2, \ldots, 8$ are placed in the 3×3 grid below, leaving exactly one blank square. Such a placement is called *okay* if in every pair of adjacent squares, either one square is blank or the difference between the two numbers is at most 2 (two squares are considered adjacent if they share a common side). If reflections, rotations, etc. of placements are considered distinct, compute the number of distinct okay placements.



T-4. An ellipse in the first quadrant is tangent to both the x-axis and y-axis. One focus is at $(3, 7)$, and the other focus is at $(d, 7)$. Compute d.

T-5. Let $A_1A_2A_3A_4A_5A_6A_7A_8$ be a regular octagon. Let $\mathbf{u}$ be the vector from A_1 to A_2 and let $\mathbf{v}$ be the vector from A_1 to A_8. The vector from A_1 to A_4 can be written as $a\mathbf{u} + b\mathbf{v}$ for a unique ordered pair of real numbers (a, b). Compute (a, b).

T-6. Compute the integer n such that $2009 < n < 3009$ and the sum of the odd positive divisors of n is 1024.

T-7. Points A, R, M, and L are consecutively the midpoints of the sides of a square whose area is 650. The coordinates of point A are $(11, 5)$. If points R, M, and L are all lattice points, and R is in Quadrant I, compute the number of possible ordered pairs (x, y) of coordinates for point R.

T-8. The *taxicab distance* between points (x_1, y_1, z_1) and (x_2, y_2, z_2) is given by

$$d((x_1, y_1, z_1), (x_2, y_2, z_2)) = |x_1 - x_2| + |y_1 - y_2| + |z_1 - z_2|.$$

The region $\mathcal{R}$ is obtained by taking the cube $\{(x, y, z) : 0 \leq x, y, z \leq 1\}$ and removing every point whose taxicab distance to any vertex of the cube is less than $\frac{3}{5}$. Compute the volume of $\mathcal{R}$.

T-9. Let a and b be real numbers such that

$$a^3 - 15a^2 + 20a - 50 = 0 \quad \text{and} \quad 8b^3 - 60b^2 - 290b + 2575 = 0.$$

Compute $a + b$.

T-10. For a positive integer n, define $s(n)$ to be the sum of n and its digits. For example, $s(2009) = 2009 + 2 + 0 + 0 + 9 = 2020$. Compute the number of elements in the set $\{s(0), s(1), s(2), \ldots, s(9999)\}$.